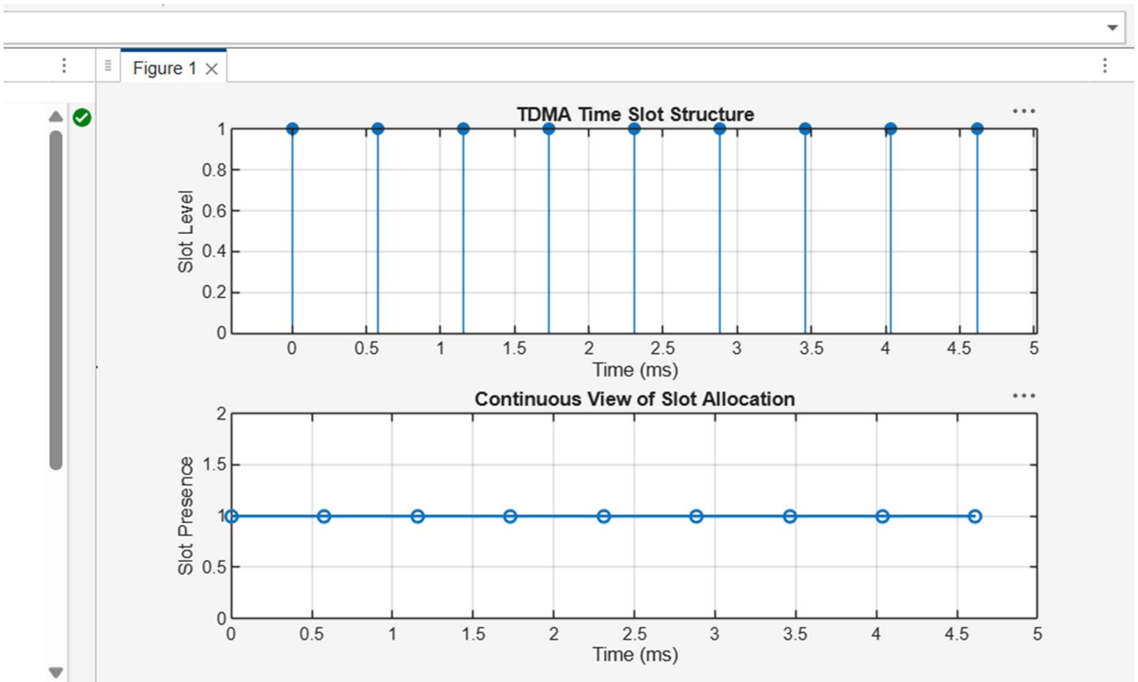
Ex.No. 2: Implementing TDMA Slot Allocation in GSM.

Objective 1

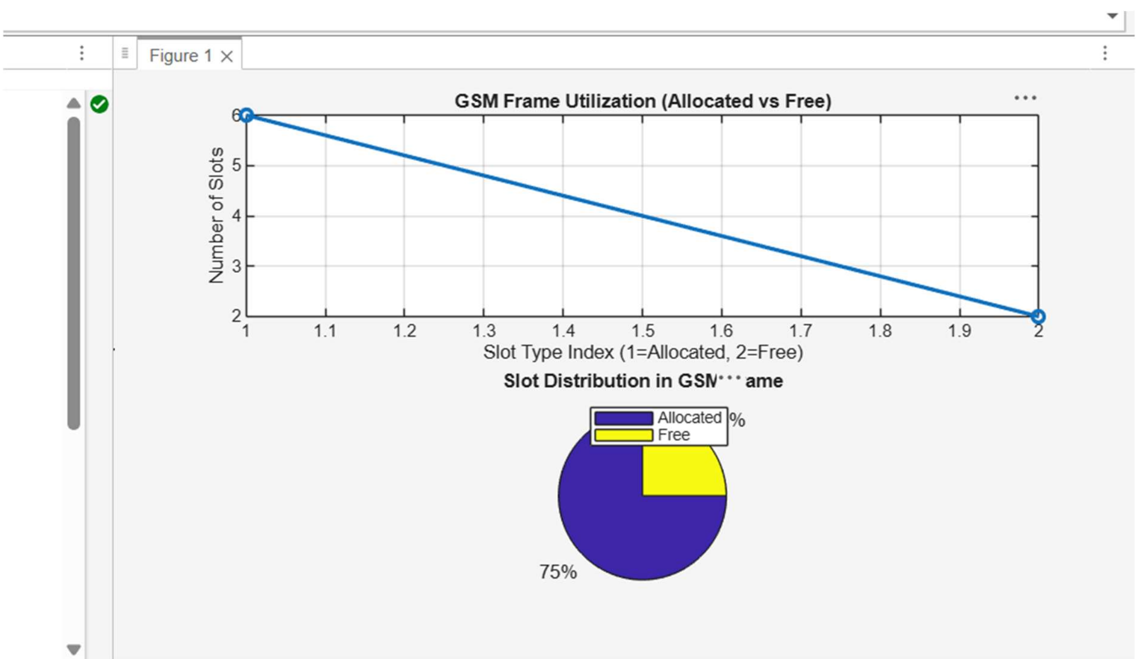
```
clc; clear; close all;
FrameTime = 4.615; Slots = 8;
SlotDuration = FrameTime/Slots;
t = 0:SlotDuration:FrameTime;
figure
subplot(2,1,1)
stem(t,ones(size(t)),'filled')
title('TDMA Time Slot Structure')
xlabel('Time (ms)')
ylabel('Slot Level')
grid on
subplot(2,1,2)
plot(t,ones(size(t)),'-o','LineWidth',1.5)
title('Continuous View of Slot Allocation')
xlabel('Time (ms)')
ylabel('Slot Presence')
grid on
```



Objective 2

```
clc; clear; close all;

TotalSlots = 8;

AllocatedSlots = 6;

FreeSlots = TotalSlots - AllocatedSlots;

Utilization = (AllocatedSlots/TotalSlots)*100;

disp('Frame Utilization (%)')

disp(Utilization)

figure

% Graph 1: Simple Line Plot (SAFE - no bar function)

subplot(2,1,1)

plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth',2)
```

```
title('GSM Frame Utilization (Allocated vs Free)')
xlabel('Slot Type Index (1=Allocated, 2=Free)')
ylabel('Number of Slots')
grid on
% Graph 2: Pie Chart (SAFE)
subplot(2,1,2)
pie([AllocatedSlots FreeSlots])
title('Slot Distribution in GSM Frame')
legend('Allocated','Free')
```



Objective 3

```
clc; clear; close all;

Users = 1:8;

TotalSlots = 5;

Alloc = zeros(1,8);

% Total users

% Available slots

% Initialize allocation

Alloc(1:TotalSlots) = 1;  % Assign slots to first users

Blocked = 1 - Alloc;

% Remaining users blocked

disp('User Wise Slot Allocation')

disp(table(Users',Alloc','VariableNames',{'User','Status(1=Allocated,0
=Blocked)'}))

figure

% Graph 1: Allocation per user

subplot(2,1,1)

stem(Users,Alloc,'filled','LineWidth',2)

title('TDMA Slot Allocation per User')

xlabel('User Index')

ylabel('Slot Status (1=Allocated, 0=Blocked)')

ylim([-0.2 1.2])

grid on

% Graph 2: System view (safe alternative to bar)
```

```
subplot(2,1,2)
plot(Users,Alloc,'-o','LineWidth',2)
hold on
plot(Users,Blocked,'-s','LineWidth',2)
title('TDMA Slot Utilization Analysis')
xlabel('User Index')
ylabel('Status')
legend('Allocated','Blocked')
grid on
```